Suggested Teaching Guidelines for

Practical Machine Learning

PGCP-AI February 2026

Duration: 60 Hours Theory + 40 hours Lab + 20 Hours Self-Learning

Objective: Practicing Machine Learning Algorithms

Prerequisites: Good knowledge of Python Programming and Statistics

Evaluation method: Theory exam– 40%
Lab Exam - 40%
Internal exam- 20%

List of Books / Other training material

Courseware:

Machine Learning using Python, Manaranjan Pradhan, U Dinesh Kumar, Wiley India, 1st Edition

Reference Book:

1. Introduction to Machine Learning with Python – A guide for Data Scientists by Andreasmuller, Sarah Guido.
2. Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2, 3rdEdition by Sebastian Raschka
3. Hands-On Machine Learning with Scikit-Learn, Keras and Tensor Flow: Concepts, Toolsand Techniques to Build Intelligent Systems by Aurelien Geron
4. AI and Machine Learning for Coders: A Programmer's Guide to Artificial Intelligence byLaurence Moroney
5. Machine Learning for Big Data: Hands- On for Developer by Jasaon Bell, Wiley

Session 1: (2T + 1SL)
Theory:

- Machine Learning in Nutshell
- Supervised and Unsupervised Learning
- ML Applications
- Evaluating ML techniques

Self-Learning:

- Reflect on why traditional rule-based systems fail in complex decision-making scenarios
- Examine real-world examples where supervised and unsupervised learning are applied differently
- Study the importance of evaluation in machine learning and consequences of choosing incorrect metrics

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Session 2 & 3: (5T + 2L + 1SL)
Theory and Lab:

- Feature engineering and Data Pre-processing: Data Preparation, Feature creation,
- Data cleaning & transformation
- Data Validation &Modelling
- Feature selection Techniques
- Dimensionality reduction

Self-Learning:

- Analyze how poor data preparation impacts model performance more than algorithm choice
- Explore real examples of feature creation improving prediction accuracy
- Study the trade-off between dimensionality reduction and information loss

Machine Learning Algorithms:
Session 4 & 5: (4T + 3L + 1SL)
Theory:

- Principal Component analysis (PCA)
- t-SNE
- K means
- Distance Measure and Data Preparation – Scaling & Weighting

Lab Assignments:

Download any dataset from kaggle.

(a) Form n no .of clusters according to your observation.
(b) Get WSS value for each cluster.
(c) Find best K value

Explore t-SNE

Self-Learning:

- Compare PCA and t-SNE from an interpretability and visualization perspective
- Reflect on how distance measures influence clustering outcomes
- Examine why choosing the correct value of K is more art than formula

Session 6: (3T + 2L + 1SL)
Theory:

- Evaluation and Profiling of Clusters
- Hierarchical Clustering
- Clustering Case Study

Lab Assignments: Implement the model hierarchical clustering on any dataset loaded from kaggle.

Self-Learning:

- Study how hierarchical clustering differs philosophically from partition-based clustering
- Analyze dendrograms as decision tools rather than visual artifacts

Session 7: (2T + 2L + 1SL)
Theory and Lab:

- DBSCAN Clustering

Self-Learning:

- Examine scenarios where density-based clustering outperforms K-Means
- Reflect on the challenges of selecting ε (epsilon) and minimum points

Session 8 & 9: (4T + 4L + 1SL)
Theory:

- Linear Regression
- Logistic Regression
- Polynomial Regression
- Stepwise Regression
- Ridge Regression
- Lasso Regression
- Elastic Net Regression

Lab Assignments:

Download Dataset, perform linear, Ridge, Lasso, Logistic, Polynomial regression and check forMAE, MSE, RMSE and also check F1-score and explain with conclusion.

Self-Learning:

- Study why regularization is required in real-world regression problems
- Compare Ridge, Lasso, and Elastic Net from a bias–variance perspective
- Reflect on situations where logistic regression is preferred over linear regression

Session 10 & 11: (4T + 3L + 1SL)
Theory:

- Support vector Machines
- Basic classification principle of SVM
- Linear and Nonlinear classification (Polynomial and Radial)

Lab Assignments:

Download Air Quality Dataset from Kaggle Predict Air Quality Index using Linear regressionand classify it into five categories using SVM (i.e. Very good, good, moderate, poor, worst)

Self-Learning:

- Visualize how margins and hyperplanes define classification boundaries
- Study why kernel methods allow non-linear separation without explicit feature expansion

Session 12: (2T + 2L + 1SL)
Theory:

- Discriminant Analysis (Linear and Quadratic)
- K-Nearest Neighbors Algorithm

Lab Assignments:
Load any dataset from kaggle, preprocess the data and apply KNN Algorithm.

Self-Learning:

- Compare distance-based learning (KNN) with probabilistic decision boundaries
- Reflect on how data distribution assumptions affect discriminant analysis

Session 13: (3T + 2L + 1SL)
Theory:

- Decision Trees
- Classification and Regression Trees

Lab Assignments:
Load any dataset from UCI Machines preprocess the data and apply Decision tree Algorithm.

Self-Learning:

- Examine why decision trees are easy to interpret yet prone to overfitting
- Reflect on how splitting criteria influence model behavior

Session 14: (2T + 2L + 1SL)
Theory:

- Bayesian analysis and Naïve Bayes classifier
- Assigning probabilities and calculating results

Lab Assignments:
Load any dataset preprocess the data and apply Naïve Bayes Algorithm.

Self-Learning:

- Analyze why Naïve Bayes performs well despite unrealistic independence assumptions
- Study real-world applications where probabilistic reasoning is preferred

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Session 15 & 16: (5T + 3L + 2SL)
Theory:

- Concept of Model Ensembling
- Random forest, Gradient boosting Machines, Model Stacking

Lab Assignments:
Implement Random Forest and gradient boosting algorithm on any dataset.

Self-Learning:

- Reflect on why ensembles outperform single models in practice
- Compare bagging, boosting, and stacking from a learning strategy perspective

Session 17 & 18: (4T + 2L + 1SL)
Theory:

- Association rules mining
- Apriori algorithms

Lab Assignments:
Generate Association Rules for shop using Apriori.

Self-Learning:

- Study how support, confidence, and lift guide business decision-making
- Examine real retail or recommendation scenarios driven by association rules

Session 19 & 20: (5T + 3L + 2SL)
Theory:

- Moving average, Exponential Smoothing
- Holt's Trend Methods
- Holt-Winters' Methods for seasonality

Lab Assignments:
Implement the smoothing models on the time series data.

Self-Learning:

- Reflect on why time-dependent data violates standard ML assumptions
- Study how trend and seasonality affect forecasting reliability

Session 21 & 22: (5T + 3L + 1SL)
Theory:

- Auto-correlation (ACF & PACF)
- Auto-regression
- Auto-regressive Models

- Moving Average Models
- ARMA &ARIMA

Lab Assignments:

Download one data set and calculate Auto correlation. Explain ARMA and ARIMA model intime series and Explain difference between them by applying on any dataset.

Self-Learning:

- Examine the intuition behind autocorrelation and partial autocorrelation
- Reflect on why ARIMA modeling requires careful interpretation

Session 23: (3T + 2L + 1SL)

Theory:

- ML in Real-Time
- Algorithm Performance Metrics
- ROC and AOC
- Confusion Metrix
- F1 Score
- MSE and MAE

Lab Assignments:

Calculate Confusion matrix, F1 score, MSE, RMSE, MAE scores on any model.
Load a dataset in Jupyter notebook and implement any classification algorithm and generateclassification report and accuracy score.

Self-Learning:

- Analyze why accuracy alone is misleading for imbalanced datasets
- Study how metric choice affects business and operational decisions

Session 24 & 25: (5T + 3L + 2SL)

Theory:

- Recommendation Systems
- Data Collection & Storage, Data Filtering
- Collaborative Filtering
- Factorization Methods
- Evaluation Metrics: Recall, Precision, RMSE, Mean Reciprocal Rank, MAP at K,NDCG

Lab Assignments:

Download Movie dataset from kaggle and implement the movie recommendation system.

Self-Learning:

- Study differences between content-based and collaborative filtering
- Reflect on ethical and bias concerns in recommendation systems

Session 26: (2T + 2L + 1SL)
Theory:

- Anomaly detection
- Point, Contextual and Collective Anomaly
- Supervised and Unsupervised anomaly detection

Lab Assignments:

Apply Anomaly Detection on dataset using isolation forest algorithm and calculate the accuracy.

Self-Learning:

- Explore Usage of ML algorithms, Algorithm performance metrics (confusion matrix sensitivity,Specificity, ROC, AOC, F1score, Precision, Recall, MSE, MAE)
- Research and Study Credit Card Fraud Analysis, Intrusion Detection system